

MINDPULSE

Intelligent Student Mental Health & Cognitive Wellness Management System

A Modular, Privacy-First Psychological Analytics & Stress Copilot

Course Name:	Problem Solving and Programming using Python
Project Domain:	Applied Health Informatics / Cognitive Computing / Software Engineering
Evaluation Platform:	VITyarthi — Build Your Own Project
Implementation Language:	Python 3.13 (Modular OOP, SQLite Persistence)
Test Suite Status:	24 / 24 Automated Unit Tests Passing (100%)
Submission Date:	September 28, 2026
Student Name / ID:	Student Submission (VITyarthi Registered Candidate)

1. Introduction

College life is exciting, but it brings intense daily pressure. Between attending morning lectures, submitting weekly lab assignments, preparing for mid-semester exams, and managing everyday life, students often struggle to maintain a balanced routine. Late-night study sessions, irregular meal timings, and high academic stress have become standard for many engineering undergraduates. Unfortunately, most students rarely notice how much their daily habits are draining them until they feel completely burnt out.

There are dozens of mood-tracking and wellness apps available on mobile app stores, but almost all of them have serious drawbacks for students. Most require costly monthly subscriptions, require constant internet connectivity, and push users to upload personal diary entries onto remote cloud servers. For students who care about privacy, sharing personal emotions and stress notes with third-party companies is unacceptable. On top of that, most commercial apps just give random motivational quotes rather than showing real connections between habits and mood.

I developed **MindPulse** to solve these problems directly on a student's personal computer. MindPulse is an offline-first, terminal-based wellness tracker written in Python. It provides students with a private, distraction-free space to record their daily habits (sleep, study, and exercise), write personal reflection journals, take standard psychological screening tests (PSS-10, PHQ-9, GAD-7), and mathematically analyze how their sleep duration and study hours directly correlate with their overall mood.

2. Problem Statement

The main problem this project tackles is that students lack a simple, trustworthy, and private tool to keep track of their daily mental wellness and study-sleep habits. Specifically, students face several recurring issues:

- **Ignoring Stress Until It Is Too Late:** Most students do not regularly reflect on their emotional state and only realize their stress levels when exam deadlines arrive and burnout sets in.
- **Privacy and Data Concerns:** Students hesitate to write honest, personal thoughts in apps that store data on remote cloud servers where it might be exposed or analyzed.
- **Not Seeing the Link Between Habits and Mood:** Many students do not realize how severely pulling an all-nighter or studying 10 hours straight affects their mood and energy over the following days.
- **Relying on Arbitrary Guesswork:** Most apps ask 'How do you feel?' with smiley faces instead of using standard screening tests recognized by counselors and psychologists.

3. Functional Requirements

As mandated by Section 2.1 of the project guidelines, MindPulse provides comprehensive functionality organized into five major decoupled modules:

Module Name	Key Functional Capabilities	Input / Output Structure
1. Authentication & Security	Multi-user account provisioning, salted cryptographic password hashing, constant-time verification, session management.	In: Username, password Out: Authenticated session context, session state token
2. Daily Habit & Wellbeing Tracker	Records mood score (1-10), mood label, energy level (1-5), sleep hours, study duration, and exercise minutes.	In: Daily numeric metrics & notes Out: Validated database record, historical trend table

3. Reflective Journal & Sentiment Engine	Textual journal recording with automated rule-based NLP sentiment scoring (-1.0 to +1.0), dominant emotion classification, and hashtag extraction.	In: Journal title & text body Out: Polarity score, classified emotion tag, extracted themes
4. Clinical Psychometrics Suite	Administers PSS-10 (Perceived Stress), PHQ-9 (Depression), and GAD-7 (Anxiety) with reverse-scored items and clinical classification.	In: Standardized item responses (0-4) Out: Raw score, severity band, clinical advice
5. Analytics & Risk Engine	Calculates Pearson correlation coefficients (\$r\$) between lifestyle parameters and computes composite Burnout Risk Index (BRI).	In: Rolling 7-to-30 day log sequences Out: Pearson \$r\$, BRI percentage, warning factors
6. Coping Recommender & Crisis Dispatch	Delivers step-by-step CBT exercises (Box Breathing, Grounding 5-4-3-2-1) and national/campus emergency hotline directories.	In: User request / risk state Out: Formatted somatic guide, emergency phone numbers

4. Non-Functional Requirements

MindPulse fulfills six critical non-functional criteria as specified in Section 2.2:

- **Security & Privacy:** Zero network egress. Passwords protected with PBKDF2-HMAC-SHA256 (100,000 iterations) with individual 16-byte cryptographically secure random salts. Timing attacks neutralized using constant-time comparison.
- **Performance:** Sub-10ms response latency for all analytical, psychometric, and data queries. In-memory caching and SQLite B-tree indices ensure instantaneous execution.
- **Usability:** Intuitive, distraction-free command-line interface featuring ANSI color coding, dynamic ASCII sparklines, and comprehensive input boundary validation.
- **Reliability & Crash-Resilience:** Built on SQLite ACID transactions with context managers ensuring automatic commit and rollback on errors, preventing data corruption.
- **Maintainability:** Strict Object-Oriented Clean Architecture following the Repository pattern, separating Domain Models, Data Access, Business Analytics, and UI controllers.
- **Resource Efficiency:** Ultra-lightweight memory footprint (< 25MB RAM), zero persistent background daemon load, ensuring optimal battery conservation on student laptops.

5. System Architecture

MindPulse is architected according to the **Clean Multi-Tiered Layered Architecture**. This pattern enforces strict separation of concerns, ensuring high cohesion, low coupling, and maximum testability.

LAYER 1: PRESENTATION & INTERACTION LAYER

src/ui/cli.py (State Controller) | src/ui/views.py (ANSI Views, ASCII Sparklines)

■ (Dispatches user actions / Receives formatted domain views)

LAYER 2: DOMAIN LOGIC & APPLICATION SERVICES

- src/security/auth.py : PBKDF2 Cryptographic Security & Session State
- src/analytics/sentiment.py : Lexicon NLP Polarity & Emotion Extraction
- src/analytics/statistics.py : Pearson Correlation (\$r\$) & Burnout Risk Modeling
- src/assessments/clinical_scales.py : PSS-10, PHQ-9, GAD-7 Psychometric Scoring
- src/assessments/recommendations.py : CBT Exercises & Emergency Directory

■ (Operates on Domain Dataclasses / Invokes Repositories)

LAYER 3: DATA ACCESS & REPOSITORY LAYER

src/storage/repository.py : UserRepository, JournalRepository, MoodRepository, AssessmentRepository

■ (Manages transactional connection lifecycles & SQL queries)

LAYER 4: PERSISTENCE & STORAGE LAYER

src/storage/database.py (DatabaseManager with ACID context scopes) ➡ **SQLite Database**
(data/mindpulse.db)

6. Design Diagrams

This section presents comprehensive structural and behavioral UML design models for the system.

6.1 Use Case Diagram

Actor	Use Case	Description
Student	Register / Authenticate	Creates account or logs in with salted PBKDF2 authentication.
Student	Log Daily Wellbeing & Habits	Inputs mood score, energy, sleep duration, study hours, and exercise.
Student	Write Reflective Journal	Submits private text; system extracts emotional valence and hashtags.
Student	Take Psychometric Screening	Answers standardized PSS-10, PHQ-9, or GAD-7 questionnaires.
Student	Evaluate Burnout Risk	Reviews rolling 7-day Burnout Risk Index (BRI) and warning alerts.
Student	Practice CBT Coping	Executes Box Breathing, 5-4-3-2-1 Grounding, or PMR somatic steps.
Student	Export Data to CSV	Exports structured CSV file for longitudinal medical or self review.

6.2 Sequence Diagram: Reflective Journaling Execution Flow

Step	Originating Entity	Receiving Entity	Operation / Message Payload
1	Student	MindPulseCLI	Submits journal title and raw reflection text
2	MindPulseCLI	SentimentAnalyzer	Invokes analyze(text) with negation & amplifier modifiers
3	SentimentAnalyzer	MindPulseCLI	Returns (polarity_score, dominant_emotion, tags)
4	MindPulseCLI	JournalRepository	Calls add_entry(journal_entry_obj)
5	JournalRepository	DatabaseManager	Executes ACID SQL INSERT within transaction()
6	DatabaseManager	JournalRepository	Commits record and returns lastrowid
7	MindPulseCLI	TerminalViews	Renders confirmation, colored emotion tag, and topical keywords

6.3 Database & Storage Design: Relational Schema & ER Mapping

The relational storage design incorporates strict referential integrity with cascading deletes and index optimizations.

Table Name	Primary Key	Foreign Keys	Attributes & Constraints
users	user_id (INT AUTO)	None	username (TEXT UNIQUE), password_hash (TEXT), salt (TEXT), created_at, last_login
journal_entries	entry_id (INT AUTO)	user_id → users.user_id	title (TEXT), content (TEXT), sentiment_score (REAL), dominant_emotion (TEXT), tags (JSON TEXT), created_at
mood_logs	log_id (INT AUTO)	user_id → users.user_id	date (TEXT), mood_score (INT 1-10), mood_label (TEXT), energy_level (INT 1-5), sleep_hours (REAL), study_hours (REAL), exercise_minutes (INT), notes (TEXT), UNIQUE(user_id, date)
assessment_results	assessment_id (INT AUTO)	user_id → users.user_id	assessment_type (TEXT), raw_score (INT), max_score (INT), severity_level (TEXT), recommendation (TEXT), completed_at

7. Design Decisions & Rationale

- **Why SQLite Instead of a Cloud Database:** SQLite is built right into Python and saves everything in a local file (mindpulse.db). This means the student does not need to set up any database server, no data ever leaves their computer, and the app works completely without internet.
- **Why a Rule-Based Sentiment Dictionary Instead of Heavy ML Models:** Loading a multi-gigabyte machine learning library like PyTorch or transformers would make the app very slow to install and heavy on RAM. A lightweight Python dictionary with negation logic runs instantaneously and requires zero extra pip packages.
- **Why Standard Clinical Tests (PSS-10, PHQ-9, GAD-7):** Rather than making up random questions, I implemented standard psychological screening tests used by counselors so students receive meaningful, recognized feedback.
- **Why the Repository Pattern:** Keeping SQL queries in dedicated repository files keeps the rest of the application clean and makes it easy to run automated tests with temporary in-memory databases.

8. Implementation Details

MindPulse is implemented in modular Python 3 across six cohesive sub-packages. Key algorithmic achievements include:

1. Burnout Risk Index (BRI) Mathematical Formulation:

The BRI computes a cumulative stress penalty (0 - 100%) by evaluating rolling 7-day moving averages of student habits: penalties are assessed for severe sleep deficit (sleep < 5.0h: +30%), excessive academic workloads (study > 8.0h: +25%), depressed mood (mood < 4.0/10: +25%), mental exhaustion (energy < 2.0/5: +15%), and sedentary habits (exercise < 10m: +10%).

2. Pearson Correlation Coefficient (\$r\$) Implementation:

Measures linear dependence between lifestyle habits and emotional vitality using the formula:
$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$
. Enables students to empirically observe how incremental hours of sleep or physical activity directly improve their mood scores.

9. Terminal Execution Results & Screenshots

The following verified execution captures illustrate the interactive command-line interface in production:

Result 1: Habit Logging & ASCII Trend Sparkline

```
===== DATE | MOOD | LABEL |
SLEEP | STUDY | EXERCISE | ENERGY
===== 2026-09-22 | 5/10 |
Stressed | 5.0h | 7.5h | 10m | 2/5 2026-09-23 | 4/10 | Overwhelmed | 4.5h | 8.0h | 0m | 2/5
2026-09-24 | 6/10 | Tired | 6.0h | 6.0h | 15m | 3/5 2026-09-25 | 7/10 | Productive | 7.0h | 5.0h |
30m | 4/5 2026-09-26 | 8/10 | Content | 7.5h | 4.0h | 40m | 4/5 2026-09-27 | 8/10 | Energetic |
8.0h | 3.5h | 45m | 5/5
===== Trend Sparkline (last
6 logs): ■■■■■ Statistical Correlation Insights (Pearson r): • Sleep vs. Mood: +0.96 (Positive
indicates more sleep improves mood) • Study vs. Mood: -0.98 (Negative suggests heavy workload
lowers mood) • Exercise vs. Mood: +0.94 (Positive shows physical activity boosts spirits) • Sleep
vs. Energy: +0.97
```

Result 2: Reflective Journaling Real-Time Sentiment Feedback

```
Write your journal entry below (Press Enter when done): > Completed the project evaluation today.
Feeling deeply relieved, peaceful, and accomplished! ✓ Journal entry preserved locally! Detected
Emotional Tone: Joy (Polarity Score: +0.92) Extracted Topical Themes: project, evaluation,
relieved, peaceful, accomplished
```

10. Testing Approach & Verification

MindPulse follows a rigorous automated testing paradigm using Python's native unittest framework. All test fixtures execute against an in-memory SQLite database (:memory:) to ensure test isolation, zero disk pollution, and sub-second execution.

Test Suite Module	Tests Executed	Verification Focus	Status
test_models.py	4	Dataclass instantiation, default timestamps, field integrity	PASS
test_security.py	4	PBKDF2 hashing, salt generation, constant-time checks, auth validation	PASS
test_storage.py	3	Schema creation, ACID transactions, CRUD operations on repositories	PASS
test_sentiment.py	5	Polarity scoring, negation handling, emotion detection, empty inputs	PASS
test_assessments.py	5	PSS-10 reverse scoring, PHQ-9 & GAD-7 scoring bands, bounds errors	PASS
test_statistics.py	3	Mean, std dev, Pearson correlation (\$r=\pm 1.0\$), Burnout Risk Index	PASS
Total Suite	24 Tests	End-to-End System & Unit Test Coverage (0.51s execution)	100% PASS

11. Challenges Faced & Practical Solutions

1. In-Memory Database Isolation During Unit Testing: When testing SQLite using ``:memory:``, opening a new connection can accidentally create a completely blank in-memory database, losing test tables. *Solution:* I designed ``DatabaseManager`` to hold onto a shared connection object during test mode so all test suites run cleanly against the same schema.

2. Handling Negation in Sentiment Analysis: A simple word counter easily misinterprets phrases like 'not happy' as positive because it sees the word 'happy'. *Solution:* I wrote a state tracking buffer that watches for words like 'not', 'no', and 'cant', and flips the sentiment value of the next word so the sentence is scored accurately.

3. Reverse-Scoring in the PSS-10 Stress Scale: Standard questionnaires like PSS-10 have positively worded questions (e.g. 'felt confident handling problems') that must be reverse-scored (4 minus response). *Solution:* I created a dedicated index set for reverse questions to ensure the final stress score matches standard clinical scoring tables.

12. Learnings & Key Takeaways

- **Modular Python Programming:** Learned how to organize a project into clean packages (``core``, ``storage``, ``analytics``, ``ui``) instead of cramming everything into one messy script.
- **Database Management with SQLite:** Gained hands-on experience creating tables, foreign keys, and using Python context managers to prevent database lockups and crashes.
- **Writing Automated Unit Tests:** Learned how to write and run 24 unit tests using Python's ``unittest`` library to verify that login, database queries, and math formulas work reliably.
- **Applied Math in Software:** Implemented Pearson correlation formulas from scratch to help students see real mathematical proof of how sleep impacts their mood.

13. Future Enhancements

- **Modern Graphical User Interface (GUI):** Develop an optional cross-platform desktop UI using PyQt6 or modern web UI using FastAPI and React.
- **Local Private LLM Integration:** Integrate quantized local large language models (via Ollama or llama.cpp) to provide interactive conversational cognitive reframing without cloud exposure.
- **Wearable Biometric Ingestion:** Implement local Bluetooth synchronization with smartbands to automatically import resting heart rate, HRV, and sleep stages.

14. References

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